Exercise Sheet 1: Quantum States and Measurements

Question 1. Which of the following are *not* density matrices?

$$A = \frac{1}{4}\begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix} \in \mathbb{C}^{2\times 2}, \qquad B = \frac{1}{2}\begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix} \in \mathbb{C}^{2\times 2},$$

$$C = \frac{1}{2}\begin{pmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \in \mathbb{C}^{4\times 4}, \quad D = \begin{pmatrix} 1 \end{pmatrix} \in \mathbb{C}^{1\times 1}.$$

Question 2. Which of the following density matrices correspond to *pure* states?

$$A = \frac{1}{4}\begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix} \in \mathbb{C}^{2\times 2}, \qquad B = \frac{1}{2}\begin{pmatrix} 1 & i \\ -i & 1 \end{pmatrix} \in \mathbb{C}^{2\times 2},$$

$$C = \frac{1}{2}\begin{pmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \in \mathbb{C}^{4\times 4}, \quad D = \begin{pmatrix} 1 \end{pmatrix} \in \mathbb{C}^{1\times 1}.$$

Question 3. Which of the following density matrices correspond to *classical* states in the standard basis?

$$A = \frac{1}{4}\begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix} \in \mathbb{C}^{2\times 2}, \qquad B = \frac{1}{2}\begin{pmatrix} 1 & i \\ -i & 1 \end{pmatrix} \in \mathbb{C}^{2\times 2},$$

$$C = \frac{1}{2}\begin{pmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \in \mathbb{C}^{4\times 4}, \quad D = \frac{1}{2}\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \in \mathbb{C}^{4\times 4}.$$

Question 4. Recall that $\mathcal{S}(\mathcal{H})$ denotes the set of density matrices on $\mathcal{H}$. For arbitrary linear operators ρ_1, ρ_2 on $\mathcal{H}$, consider the statements:

(I) $\rho_1 \in \mathcal{S}(\mathcal{H})$ and $\rho_2 \in \mathcal{S}(\mathcal{H})$.
(II) $\rho_1\rho_2 \in \mathcal{S}(\mathcal{H})$.

Which of the following holds in general?

(A) (I) $\implies$ (II).
(B) (I) $\impliedby$ (II).
(C) (I) $\iff$ (II).
(D) None of those.

Question 5. Consider the pure qubit state

$$|\psi\rangle = \frac{\sqrt{3}}{2}|0\rangle + \frac{1}{2}|1\rangle.$$

(a) Write its density matrix $\rho = |\psi\rangle\langle\psi|$ in the standard basis.
(b) Find the outcome probabilities when measuring in the basis $\{|0\rangle, |1\rangle\}$.
(c) Find the outcome probabilities when measuring in the basis $\{|+\rangle, |-\rangle\}$, where $|\pm\rangle = (|0\rangle \pm |1\rangle)/\sqrt{2}$.

Question 6. Which of the following pairs $(\mu(0), \mu(1))$ define a POVM on $\mathbb{C}^2$? Which of the valid POVMs are projective? Here I is the 2×2 identity matrix.

$$\text{(A)} \quad \left(|0\rangle\langle 0|,\ |1\rangle\langle 1|\right), \quad \text{(B)} \quad \left(\frac{1}{2}I,\ \frac{1}{2}I\right),$$

$$\text{(C)} \quad \left(\frac{1}{2}I,\ \frac{1}{4}I\right), \qquad \text{(D)} \quad \left(\begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix},\ \begin{pmatrix}0 & 0\\ 0 & 2\end{pmatrix}\right).$$

For each valid POVM, compute the outcome probabilities for $\rho = \frac{1}{4}\begin{pmatrix}3 & 0\\ 0 & 1\end{pmatrix}$.

Question 7. Which of the following vectors are Bloch vectors of qubit states? Which of the valid vectors describe pure states, and which describe mixed states?

$$r_A = (0, 0, 1), \qquad r_B = \left(\frac{1}{2}, \frac{1}{2}, 0\right), \qquad r_C = (1, 1, 0), \qquad r_D = (0, 0, 0).$$

For each valid vector, find the probability of outcome 0 in a Z-basis measurement and the probability of outcome $+$ in an X-basis measurement.

Question 8. Let

$$\rho = \frac{1}{4}\begin{pmatrix}3 & 1\\ 1 & 1\end{pmatrix}, \qquad O = 2|0\rangle\langle 0| - |1\rangle\langle 1|.$$

(a) What are the possible outcomes when measuring the observable O, and what are their probabilities in the state ρ?
(b) Compute $\mathbb{E}_\rho[O]$ both as a weighted average of the outcomes and using $\mathrm{tr}[\rho O]$.
(c) Can a single measurement of O return the value $\mathbb{E}_\rho[O]$? Explain what this expectation value means.

Solutions

1. **B only.** It is not Hermitian: the off-diagonal entries are not complex conjugates. A, C, and D are positive semidefinite and have trace one.

2. **B, C, and D.** They have rank one (equivalently, $\rho^2 = \rho$). A has eigenvalues $3/4$ and $1/4$, so it is mixed.

3. **A and D.** Classical states are diagonal in the specified basis; their diagonal entries give the probability distribution.

4. **D: neither implication holds.** For (I)$\not\Rightarrow$(II), take $\rho_1 = \rho_2 = I_2/2$: their product $I_2/4$ has trace $1/2$. For (II)$\not\Rightarrow$(I), take $\rho_1 = 2P$, $\rho_2 = P/2$ with P a rank-one orthogonal projection: the product is the state P, but neither factor has trace one.

5. **(a)** $\rho = \frac{1}{4}\begin{pmatrix} 3 & \sqrt{3} \\ \sqrt{3} & 1 \end{pmatrix}$, by taking the outer product. **(b)** $(p(0), p(1)) = (3/4, 1/4)$. **(c)** $p(\pm) = (2 \pm \sqrt{3})/4$. In either basis, the Born rule gives $p(x) = |\langle x|\psi\rangle|^2$.

6. **A and B are POVMs; only A is projective.** Their effects are positive and sum to I_2. C sums to $3I_2/4$; D has a negative eigenvalue in its first effect. B is not projective since $(I_2/2)^2 \neq I_2/2$. Using $p(x) = \mathrm{tr}[\rho\mu(x)]$, A gives $(3/4, 1/4)$ and B gives $(1/2, 1/2)$.

7. **A is pure; B and D are mixed; C is invalid.** A Bloch vector is valid iff $\|r\| \leq 1$, and pure iff $\|r\| = 1$. The lengths are $1, 1/\sqrt{2}, \sqrt{2}, 0$, respectively. Since $p_Z(0) = (1+z)/2$ and $p_X(+) = (1+x)/2$,

$$(p_Z(0), p_X(+)): \qquad A:\ (1, 1/2), \quad B:\ (1/2, 3/4), \quad D:\ (1/2, 1/2).$$

8. **(a)** Outcomes 2 and -1 have probabilities $3/4$ and $1/4$, from the spectral projections of O. **(b)** $\mathbb{E}_\rho[O] = 2(3/4) - (1/4) = 5/4$; also $\mathrm{tr}[\rho O] = \frac{1}{4}(6-1) = 5/4$. **(c)** No: $5/4$ is not an eigenvalue of O. It is the average approached over many independent measurements on systems prepared in state ρ.